MOSTOFA HABIB FARDIN

Undergraduate Student, Department of Industrial and Production Engineering, Bangladesh University of Engineering and Technology

Address: 18 Khaja Dewan 1st Lane, Lalbagh, Dhaka-1211
Phone: +880 1812997207
Email: mostofahabibfardin08@gmail.com
LinkedIn: linkedin.com/in/mostofahabibfardin/
Website: sites.google.com/view/mostofa-habib-fardin/home

SUMMARY

Final-year Industrial and Production Engineering student at BUET with hands-on experience in operations management, quality improvement, supply chain management and product design through industrial attachment and coursework. Recognized problem solving skills through business and case competitions (HULT, Water Hackathon), building scalable, tech-enabled solutions. Skilled in data analysis, CAD and IoT systems, with strong teamwork and leadership from high-ownership project work. Interested in applying lean thinking and structured problem-solving to deliver measurable improvements.

EDUCATION

B.Sc. in Industrial and Production Engineering *Tentative Graduation- May, 2026*
Bangladesh University of Engineering and Technology (BUET) | CGPA- **3.87 out of 4.00** (up to 7th semester)

Higher Secondary Certificate (HSC) *2020*
Notre Dame College, Dhaka | GPA- **5.00 out of 5.00**

WORK EXPERIENCE

RANCON Auto Industries Ltd. & RANCON Motorbikes Ltd. (Industrial Attachment) *Nov. 2025-Dec 2025*

- Worked in the Production Line and QC Department, assisting with **time study, tool listing, SOP upgrades and Rack ID system implementation.**
- **Identified bottlenecks** in production and practices causing **quality and safety issues,** providing impactful **recommendations** for improvement.
- Contributed to **5S, Kaizen** and **safety** enhancements, aligning operations with Japanese OEM standards.
- **Presented findings** to the Executive Director and followed up with the Production Manager to discuss potential implementation.
- Gained practical experience in **Operations Management, Quality Management** and **Lean Manufacturing.**

ACHIEVEMENTS & CERTIFICATIONS

- **1st Runner-Up, HULT Prize at BUET (2022):** Co-developed Biztarter, a platform to connect entrepreneurs, investors and professionals to foster startups.
- **Dean's List, BUET (2022, 2023):** Recognized for academic excellence.
- **60% Scholarship Winner, ISCEA Global Supply Chain Case Competition (2025):** Developed a solution using AI, Blockchain, and IoT for supply chain challenges.
- **Finalist, Water Hackathon-2025(IRC-WASH, BRAC):** Co-developed a water purification system providing safe drinking water to ~1,500 people in coastal Bangladesh.
- **Advanced MS Word, Excel, and PowerPoint for Engineers, AIPE, BUET:** Completed an advanced Microsoft Office Suite application course for engineers.
- **Perfect Attendance, Notre Dame College:** Awarded for flawless attendance.
- **Lean Six Sigma White Belt, Council for Six Sigma Certification:** Certified in process improvement and lean methodologies.

SKILLS

- **Technical Skills:** SolidWorks, CATIA, AutoCAD, ANSYS, Programming (C, Python), MATLAB, MS Office Suite, LaTeX
- **Soft Skills:** Problem-solving, Teamwork, Leadership, Critical Thinking, Time Management, Coordination, Negotiation

PROJECTS

HydroPedal

- Designed a **pedal-powered water pump** for off-grid rural households as a low-cost alternative to motorized pumps and tube-wells.
- Modelled and validated the design in **SolidWorks and ANSYS,** using a chain-sprocket drive, optimized gear ratio, and mild-steel frame for **durability and local manufacturability.**
- Delivered **~25–30 L/min at ~20 ft head,** supporting irrigation and household use with **zero operating (fuel/ electricity) cost.**

Smart IoT-Based Ammonia and Ambient Condition Monitoring System

- Built an IoT-based monitoring and control system to detect **ammonia (NH_3)** concentration and **ambient conditions** for **fertilizer-plant** safety use cases.
- Integrated MQ137 and DHT22 with ESP-01S to log real-time data, with **server-based thresholds** (30 ppm NH_3, 30°C) triggering fan and buzzer alerts.
- Developed a web dashboard for **data visualization** and reporting, improving response time and supporting safety compliance.

REFERENCES

Dr. Sultana Parveen
Professor, Department of Industrial and Production Engineering,
Bangladesh University of Engineering and Technology
Email: sparveen@ipe.buet.ac.bd

Fayjur Anam
Manager, Rancon Auto Industries Ltd.
Mobile: +880 1720828103
Email: fayjur@rancon.com.bd